4.2.38.2 Protocol

a) Report node FC-4 Types (RNFT) Request sequence

b) LS_ACC or LS_RJT Reply Sequence

4.2.38.3 Request Sequence

Addressing: The S_ID field designates the source Nx_Port requesting the FC-4 Types information. The D_ID field designates the destination Nx_Port to which the request is being made.

Payload: The format of the request Payload is shown in table 104.

Table 104 – RNFT Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	RNFT (63h)	Reserved	Maximum Size	
1	Reserved			Index

a) **Maximum Size:** Bytes 2-3 of Word 0 contain a 16-bit value that specifies the maximum length of the RNFT LS_ACC that the originator is able to accept. The value zero implies the RNFT LS_ACC may be any size.

b) **Index:** Byte 3 of Word 1 contains an 8-bit value that specifies the index of the first FC-4 Entry to be returned in the RNFT Reply.

Each FC-4 protocol supported by the responder has an index in the range from zero to (List Length – 1) that should be used to specify a subset of the entries when the entire list does not fit into one reply. NOTE - The index of the entry for a particular FC-4 TYPE may not be consistent between subsequent RNFT Requests (e.g., due to additions or deletions of supported FC-4 TYPEs).

4.2.38.4 Reply Sequence

LS_RJT: LS_RJT signifies rejection of the RNFT command.

LS_ACC: LS_ACC Signifies that the destination Nx_Port has transmitted the requested data. The format of the LS_ACC Payload is shown in table 105.

Table 105 – RNFT LS_ACC Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	02h	Reserved	Payload Length (M)	
1	Reserved	List Length	Reserved	Index
2	FC-4 Entry 1			
..	...			
N+1	FC-4 Entry N			

a) **Payload Length:** Bytes 2-3 of Word 0 contain a 16-bit value that specifies the length M of the RNFT LS_ACC Payload in bytes.

M = 8 + N*4

where

N is the number of FC-4 Entries contained in the Payload.

b) **List Length:** Byte 1 of Word 1 contains an 8-bit value that specifies the total number of FC-4 protocols supported by the responder.

If List Length exceeds Index+N then the originator may request additional records with another RNFT in which Index is increased by N.

c) **Index:** Byte 3 of Word 1 contains an 8-bit value that specifies the index of the first FC-4 Entry returned in the RNFT reply.

d) **FC-4 Entry:** The FC-4 Entry record contains a FC-4 Entry and is shown in table 106.

Table 106 – RNFT FC-4 Entry

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
1	FC-4 Type	FC-4 Qualifier		

A) **FC-4 Type:** The FC-4 TYPE code of a FC-4 protocol that is supported by the sending Nx_Port. The values are defined in FC-FS-2**.**

B) **FC-4 Qualifier:** The FC-4 Qualifier may be used to distinguish between two protocols that use the same FC-4 TYPE code.

For FC-4 type codes that are reserved or assigned for specific use in this standard {CWC: Should this be 00h to DFh?} (00h - 68h), the value of the FC-4 Qualifier shall be zero.

For Vendor specific FC-4 TYPE codes (E0h through FFh), the FC-4 Qualifier shall be selected from one of the 24-bit Company_ID values assigned by the IEEE Registration Authority to the organization that defines the Vendor specific FC-4 protocol, and that Company_ID shall be used to qualify that FC-4 TYPE in all implementations. It is up to the organization that defines the Vendor specific FC-4 protocol to assure that the protocol has a unique qualified FC-4 Type.

4.2.39 Scan Remote Loop (SRL)

4.2.39.1 Description

The SRL ELS shall require a switch to scan attached loops to determine if any L_Ports have been disabled or removed. If the switch determines any L_Ports that are currently logged in with the Fabric have been removed or disabled it shall update the name server and send an RSCN to all registered Nx_Ports.

The SRL Payload indicates whether the switch shall scan all attached loops or a single loop. If a single loop is to be scanned the Payload shall contain the FL_Port_ID of the loop to be scanned.

4.2.39.2 Protocol

a) Scan Remote Loop Request Sequence

b) LS_ACC or LS_RJT Reply Sequence

4.2.39.3 Request Sequence

Addressing: The S_ID field designates the source Nx_Port requesting Scan Remote Loop. The D_ID field designates the destination of the address identifier for the Domain Controller of the switch for which loops are being scanned. The format of the Domain Controller address is FFFCh || Domain_ID. Domain_ID is the Domain_ID of the switch being queried.

Payload: The format of the Payload is shown in table 107.

Table 107 – SRL Payload

Bits / Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	SRL (7Bh)	Reserved		
1	Flag	Flag Parameter		

a) **Flag:** Byte 0 of word 1 indicates if the FL_Port shall be scanned, if all FL_Ports within the domain shall be scanned, or the scan period that all FL_Ports within the domain shall be scanned. The meaning of bits 0-7 is given in Table 108.

Table 108 – Flag field definitions

Value (hex)	Meaning	Flag Parameter
00	All the FL_Ports within the domain shall be scanned.	Ignored
01	Only the loop attached to the FL_Port addressed in the address identifier of the FL_Port field shall be scanned.	Address identifier of the FL_Port
02	Enable periodic scanning for all FL_ports.	Scan period[a]
03	Disable periodic scanning for all FL_ports.	Ignored
All Others	Reserved	

[a] Scan period in seconds. If the scan period is set to zero the scan period is vendor specific. If the switch does not support this option it shall reject the SRL ELS with a reason code of "Unable to perform command request" and a reason code explanation of "Periodic Scanning not supported". If the switch does not support the selected value it shall reject the SRL ELS with a reason code of "Unable to perform command request" and a reason code explanation of "Periodic Scan Value not allowed".

b) **Flag Parameter:** See Table 108.

4.2.39.4 Reply Sequence

LS_RJT: LS_RJT signifies the rejection of the SRL Request

LS_ACC: LS_ACC signifies acceptance of the SRL Request. If the period scanning is enabled then the switch shall return the value of the periodic scanning period in the LS_ACC payloads Scan Period field, If the periodic scanning period is disabled then the switch shall set the LS_ACC payload Scan Period field to zero.

Table 109 – SRL LS_ACC Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	02h	Scan Period		

4.2.40 Set Bit-error Reporting Parameters (SBRP)

4.2.40.1 Description

Set SBRP ELS is used to communicate a set of bit error reporting parameters to a Port or to all Ports in a particular Domain in a Fabric. There are 3 parameters, Error Interval, Error Window, and Error Threshold. Error Interval is the time period over which bit error bursts are integrated to produce a single reported error. An Error Window is composed of one or more Error Intervals. The Error Interval Count is the number of Error Intervals occurring in an Error Window. If the Error Interval Count is greater than or equal to the Error Thrshold, a Registered Link Incident Report (RLIR) is generated with an Incident Code specifying Bit-error-rate threshold exceeded (see FC-FS-2). At the end of the Error Window, the count is set to zero and the process is repeated. See figure 2 for illustration of the parameters.

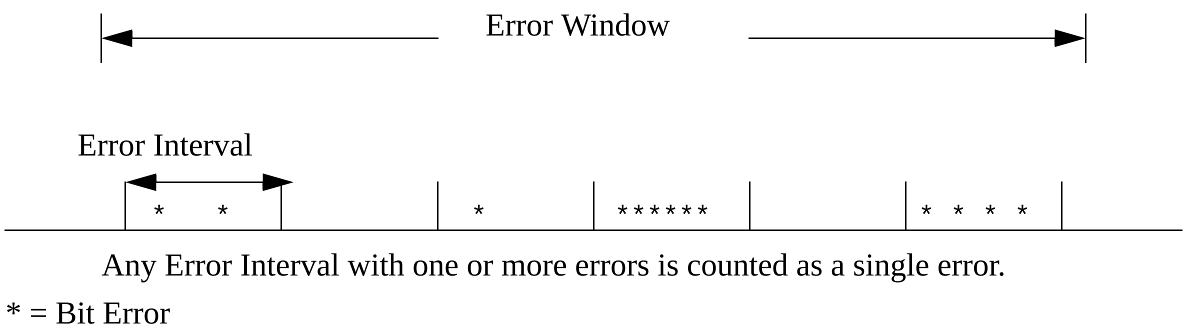

Figure 2 – Illustration of parameters

SBRP may be used to determine an acceptable set or the current set of bit error reporting parameters in the destination Port or Domain, without changing the settings.

The setting of parameters in a particular Port or Domain is done by a controlling entity that is outside the scope of this standard.

4.2.40.2 Protocol

a) Set Bit-error Reporting Parameters (SBRP) Request Sequence

b) LS_ACC or LS_RJT Reply Sequence

4.2.40.3 Request Sequence

Addressing: The S_ID designates the source Nx_Port requesting the bit error rate reporting parameters. The D_ID designates the destination Nx_Port or the Domain Controller (FFFCh || <Domain_ID>). Domain_ID is the Domain_ID of the recipient switch) to process the SBRP ELS.

Payload: The format of the SBRP Payload is shown in table 110.

Table 110 – SBRP Payload

Bits Word	31 .. 28	27 .. 24	23 .. 16	15 .. 12	11 .. 08	07 .. 03	02 .. 00
0	SBRP (7Ch)		00h	00h		00h	
1	Error Flags						
2	Error Window exponent	Error Window value		Error Interval exponent	Error Interval value		
3	Error Threshold						

a) **Error Flags:** The following bits are mutually exclusive and only one bit shall be set for each instance of the SBRP ELS.

   Bit 0 – Set Error Reporting Parameters: The bit is set to request that the destination set the Error Window, the Error Interval and the Bit-error Threshold parameters. If the destination is the Domain Controller then the request is for the switch to set all ports to the requested values.

   Bit 1 – Report Error Reporting Parameters: The bit is set to request that the destination return the active parameters, currently being enforced.

   Bits 2-31: Reserved

b) **Error Window exponent and Error Window value:** The Error Window is a time duration described by a 16-bit value. Twelve bits, Bits 28-16, are used for the base value. Four bits, Bits 31-28 are encoded to form the exponent. The product of the base value and the exponent yields the time duration in seconds. The encoded exponent is defined as follows:

   0h represents 10^{0}

   1h represents 10^{-1}

   2h represents 10^{-2}

   .
   .
   .

   Fh represents 10^{-15}

   (e.g., a base value of 300h multiplied by an exponent of 0h would yield a value of 1 times 10^{-0} or 300h seconds). The tolerance for the Error Window -0 to + 1 Error Interval.

c) **Error Interval exponent and Error Interval value:** The Error Interval is a time duration in seconds and has the same, exact definitions for exponent and base value as Error Window (e.g., a base value of 15 multiplied by an exponent of 1h would yield a value of 15 times 10^{-1} ot 1.5 seconds). The Error Interval has a tolerance of – 50% to + 50%.

d) **Error Threshold:** The Error Threshold fields specifies the basis for a comparison value with the Error Interval Count.

4.2.40.4 Reply Sequence

LS_RJT: LS_RJT signifies the rejection of the SBRP Request.

LS_ACC: LS_ACC signifies acceptance of the SBRP Request and presents SBRP data. The format of the LS_ACC Payload is shown in table 111.

The LS_ACC Payload conveys either:

a) the current Bit Error Rate Reporting Parameters (i.e., the requestor is querying for currently set parameters) or,

b) the accepted Bit Error Rate Reporting Parameters (a request to set the parameters was accepted) or,

c) acceptable Bit Error Rate Report Parameters (a request to set the parameters was rejected because the parameters requested are not within the range supported by the port or switch).

Table 111 – SBRP LS_ACC Payload

Bits / Word	31	30 .. 28	27 .. 24	23 .. 16	15 .. 12	11 .. 08	07 .. 00
0	02h			00h	00h		00h
1	SBRP Request Accepted	Reserved					
2	Error Window exponent		Error Window value		Error Interval exponent	Error Interval value	
3	Error Threshold						

a) **SBRP Request Accepted:** When a request to set the reporting parameters is indicated (request Payload word 1 bit 0 is set to one) and the recipient accepts the requested parameters, it shall set the SBRP Request Accepted bit to zero (Accepted) and echo the requested Error Window, Interval and Threshold parameters in words 2 and 3.

When a request to set the reporting parameters is indicated (i.e., request Payload word 1 bit 0 is set to one) and the recipient does not accept the parameters as requested the recipient shall set the SBRP Request Accepted bit to one and echo in words 2 and 3 those parameters that are accepted. Values not accepted by the recipient shall be set to zero.

When a request to report the parameters is indicated (request Payload word 1 bit 1 is set to one) the recipient shall set the SBRP Request Accepted bit to zero (Accepted) report the current Bit Error Rate reporting parameters being enforced. If no Bit Error Rate reporting is being enforced, words 2 and 3 shall be set to zero.

4.2.41 Report Port Speed Capabilities (RPSC)

4.2.41.1 Description

The RPSC ELS shall provide a method for a Port to report its current and potential link operating speeds. The normal response to an RPSC ELS Sequence shall be a LS_ACC ELS Sequence. If the recipient Port does not support the RPSC ELS, it shall reply with an LS_RJT ELS Sequence with a reason code of "Command not supported".

4.2.41.2 Protocol

a) RPSC Request Sequence

b) LS_ACC or LS_RJT Reply Sequence

4.2.41.3 Request Sequence

Addressing: The S_ID designates the source port that is requesting port speeds. The D_ID field designates the recipient that is to process the RPSC request. If the recipient is a Domain Controller, the request is for all the ports within that domain.

Payload: The format of the RPSC Payload is shown in table 112.

Table 112 – RPSC Payload

Bits / Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	RPSC (7Dh)	00h	00h	00h

4.2.41.4 Reply Sequence

LS_RJT: LS_RJT signifies the rejection of the RPSC Request

LS_ACC: LS_ACC signifies acceptance of the RPSC Request and its Accept Payload. The format of the LS_ACC Payload is shown in table 113.

Table 113 – RPSC LS_ACC Payload

Bits / Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	02h	Reserved	Number of entries	
1	Port Speed Capabilities (Port 1)		Port Operating Speed (Port 1)	
2	Port Speed Capabilities (Port 2)		Port Operating Speed (Port 2)	
..	...			
n	Port Speed Capabilities (Port n)		Port Operating Speed (Port n)	

a) **Number of Entries:** Specifies the number of Port entries in the Payload.

b) **Port Speed Capabilities:** Identifies the operating speed capabilities of the port

Bit 31 – 1 Gb capable
Bit 30 – 2 Gb capable
Bit 29 – 4 Gb capable
Bit 28 – 10 Gb capable
Bit 27 – 8 Gb capable
Bit 26 – 16 Gb capable
Bits 25 through 17 – reserved
Bit 16 - Unknown

c) **Port Operating Speed:** Identifies the current operating speed if set.

Bit 15 – 1 Gb Operation
Bit 14 – 2 Gb Operation
Bit 13 – 4 Gb Operation
Bit 12 – 10 Gb Operation
Bit 11 – 8 Gb Operation
Bit 10 – 16 Gb Operation
Bits 9 through 2 – reserved
Bit 1 - Unknown
Bit 0 – Speed not established.

4.2.42 Read Exchange Concise (REC)

4.2.42.1 Description

This ELS shall be used only for purposes specific to an FC-4. The REC (Read Exchange Concise) Extended Link Service requests an Nx_Port to return Exchange information for the RX_ID and OX_ID originated by the S_ID specified in the Payload of the request Sequence.

A Read Exchange Concise Request shall only be accepted if the Originator Nx_Port N_Port_ID or the Responder Nx_Port N_Port_ID of the target Exchange is the same as the N_Port_ID of the Nx_Port that makes the request. If the REC Request is not accepted, an LS_RJT with reason code "Unable to perform command request" and reason code explanation "Invalid Originator S_ID" shall be returned.

The specification of OX_ID and RX_ID shall be provided for the destination Nx_Port to locate the status information requested. A Responder destination Nx_Port shall use the RX_ID and verify that the OX_ID is consistent, unless the RX_ID is unassigned (i.e., RX_ID = FFFFh). If the RX_ID is unassigned in the request, the Responder shall identify the Exchange by means of the S_ID specified in the Payload of the request Sequence and OX_ID. An Originator Nx_Port shall use the OX_ID and verify that the RX_ID is consistent.

If the destination Nx_Port of the REC request determines that the Originator S_ID, OX_ID, or RX_ID are inconsistent, then it shall reply with an LS_RJT Sequence with a reason code of "Unable to perform command request" and a reason code explanation of "Invalid OX_ID-RX_ID combination".

The value of the Parameter field in the frame header of an REC ELS and an LS_ACC in response to an REC ELS shall be specified by the FC-4 that sends the frame. The Relative offset present bit in the frame header of an REC ELS or an LS_ACC in response to an REC ELS shall be set to zero.

4.2.42.2 Protocol

a) Read Exchange Concise (REC) Request Sequence
b) LS_ACC or LS_RJT Reply Sequence

4.2.42.3 Request Sequence

Addressing: The S_ID designates the source port that is requesting exchange information. The D_ID field designates the recipient that is requested to provide exchange information.

Payload: The format of the REC Payload is shown in table 114.

Table 114 – REC Payload

Bits Word	31 .. 24 Byte 0	23 .. 16 Byte 1	15 .. 08 Byte 2	07 .. 00 Byte 3
0	REC (13h)	00h	00h	00h
1	Reserved	Exchange Originator S_ID		
2	OX_ID		RX_ID	

Exchange Originator S_ID: shall be set to the address identifier of the target Exchange Originator.

OX_ID: shall be set to the Originator Exchange_ID value of the target Exchange.

RX_ID: shall be set to the Responder Exchange_ID value of the target Exchange.

4.2.42.4 Reply Sequence

LS_RJT: LS_RJT signifies the rejection of the REC Request

LS_ACC: LS_ACC signifies acceptance of the REC Request and returns the requested exchange information. The format of the LS_ACC Payload is shown in table 115.

Table 115 – REC LS_ACC Payload

Bits Word	31 .. 24 Byte 0	23 .. 16 Byte 1	15 .. 08 Byte 2	07 .. 00 Byte 3
0	LS_ACC (02h)	00h	00h	00h
1	OX_ID		RX_ID	
2	Reserved	Originator Address Identifier		
3	Reserved	Responder Address Identifier		
4	FC4VALUE			
5	E_STAT			

The Originator Address Identifier field shall be set to the address identifier of the Originator of the exchange about which information was requested.

The Responder Address Identifier field shall be set the address identifier of the Responder of the exchange about which information was requested.

The value of the FC4VALUE field shall be specified by the FC-4 that sends the LS_ACC to a REC ELS.

E_STAT shall be as defined in FC-FS-2 for the E_STAT field in the Exchange Status Block for the exchange about which information was requested. The bits specifying whether the Exchange is complete (Bit 29) and whether the responder holds Sequence Initiative (Bit 30) shall be valid; the setting of other bits may not be valid.

4.2.43 Exchange Virtual Fabrics Parameters (EVFP)

4.2.43.1 EVFP Messages Structure

4.2.43.1.1 EVFP Request Sequence

Protocol: Exchange Virtual Fabrics Parameters (EVFP) Request Sequence

Format: FT–1

Addressing: For an N_Port, the S_ID field shall be set to FFFFF0h, indicating the N_Port Controller of the originating N_Port. The D_ID field shall be set to FFFFFEh, indicating the F_Port Controller of the destination F_Port. For an F_Port, the S_ID field shall be set to FFFFFEh, indicating the F_Port Controller of the originating F_Port. The D_ID field shall be set to FFFFF0h, indicating the N_Port Controller of the destination N_Port

Payload: Two types of EVFP messages are defined. All EVFP Request messages share the same message structure, shown in table 116.

Table 116 – EVFP Request Payload

<table>
<tr><th>Bits
Word</th><th>31 .. 24</th><th>23 .. 16</th><th>15 .. 08</th><th>07 .. 00</th></tr>
<tr><td>0</td><td>7Fh</td><td>00h</td><td>00h</td><td>00h</td></tr>
<tr><td>1</td><td>Protocol Version</td><td>Message Code</td><td colspan="2">Transaction Identifier</td></tr>
<tr><td>2</td><td>MSB</td><td colspan="2" rowspan="2">Core N_Port_Name /
Core Switch_Name</td><td></td></tr>
<tr><td>3</td><td></td><td>LSB</td></tr>
<tr><td>4</td><td colspan="2">Reserved</td><td colspan="2">Message Payload Length</td></tr>
<tr><td>5</td><td colspan="4" rowspan="3">Message Payload</td></tr>
<tr><td>..</td></tr>
<tr><td>N</td></tr>
</table>

Protocol Version: Shall be set to one.

EVFP Message Code: Specifies the EVFP message that is to be transmitted from the source to the destination. The defined EVFP message codes are shown in table 117.

Table 117 – EVFP Message Codes

Value	Description	Reference
01h	EVFP_SYNC	4.2.43.2
02h	EVFP_COMMIT	4.2.43.3
all others	Reserved	

Transaction Identifier: Uniquely identifies an EVFP transaction between two entities. The Transaction Identifier shall be set by the EVFP Initiator, and each subsequent EVFP message shall contain the same value, until the EVFP transaction is completed.

NOTE 7 – The usage of the Transaction Identifier is very similar to the usage of an OX_ID when an Exchange Originator is enforcing uniqueness via the OX_ID mechanism (see FC-FS-2), but it is not related in any way to the OX_ID present in the Fibre Channel frames carrying the EVFP messages.

Core N_Port_Name / Core Switch_Name: If the originating FC_Port is an N_Port, this field shall be set to its Core N_Port_Name. If the originating FC_Port is an F_Port, this field shall be set to the Core Switch_Name of the Switch it belongs to.

Payload Length: Shall be set to the total length in bytes of the EVFP Payload (i.e., 20 + the Message Payload length).

4.2.43.1.2 EVFP Reply Sequence

Accept (LS_ACC)
Signifies acceptance of the EVFP request.

Accept Payload: All EVFP Accept messages share the same message structure, shown in table 118.

Table 118 – EVFP Accept Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	02h	00h	00h	00h
1	Protocol Version	Message Code	Transaction Identifier	
2	MSB	Core N_Port_Name / Core Switch_Name		
3				LSB
4	Reserved		Message Payload Length	
5	Message Payload			
..				
N				

The fields in table 118 are the same as defined in table 116.

Service Reject (LS_RJT)
Signifies the rejection of the EVFP request

Table 119 shows the use of reason codes and reason code explanations under some error conditions.

Table 119 – LS_RJT Reason Codes for EVFP

Error Condition	Reason Code	Reason Code Explanation
EVFP ELS not supported	Command not supported	No additional explanation
EVFP collision	Command already in progress	No additional explanation
Protocol Version not supported	Protocol error	No additional explanation
EVFP_COMMIT before EVFP_SYNC	Logical error	No additional explanation
Insufficient Resources	Unable to perform command request	No additional explanation
Invalid Payload Message	Protocol error	No additional explanation

4.2.43.2 EVFP_SYNC Message Payload

4.2.43.2.1 Overview

The EVFP_SYNC Message Payload carries a list of descriptors. Each descriptor is self-identifying (see table 121). The format of the EVFP_SYNC Message Payload is shown in table 120. This Message Payload is used in both EVFP_SYNC Request and EVFP_SYNC Accept.

Table 120 – EVFP_SYNC Message Payload

Bits / Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	Descriptor #1 = Tagging Administrative Status (see 4.2.43.2.2)[a]			
1				
2	Descriptor #2 = Port VF_ID (see 4.2.43.2.3)[b]			
3				
4	Descriptor #3 = Locally-Enabled VF_ID List (see 4.2.43.2.4)[c]			
..				
132				
...	...			
H				
...	Descriptor #m			
K				

[a] Decriptor #1 is required to be present in EVFP_SYNC request.
[b] Decriptor #2 is required to be present in EVFP_SYNC request.
[c] Decriptor #3 is required to be present in EVFP_SYNC request.

All descriptors share the same format, shown in table 121.

Table 121 – Descriptor Format

Bits / Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	Descriptor Control	Descriptor Type	Descriptor Length	
1	Descriptor Value			
..				
M				

Descriptor Control: Specifies the behavior of the receiving entity if the descriptor is unsupported. The defined codes are shown in table 122.

Table 122 – Descriptor Control Codes

Value	Description
01h	Critical. Abort the EVFP transaction if the descriptor is unsupported.[a]
02h	Non critical. Skip the descriptor if unsupported and continue the EVFP transaction.[a]
all others	Reserved

[a] The Descriptor Control provides extensibility to the protocol. An implementation supporting a subset of the descriptors is able to process the unknown ones as specified by the Descriptor Control value.

Descriptor Type: Specifies the type of the descriptor. The defined descriptors are summarized in table 123.

Table 123 – Descriptor Types

Value	Description	Reference
01h	Tagging Administrative Status Descriptor	4.2.43.2.2
02h	Port VF_ID Descriptor	4.2.43.2.3
03h	Locally-Enabled VF_ID List Descriptor	4.2.43.2.4
F0h .. FEh	Vendor Specific Descriptor	4.2.43.2.5
all others	Reserved	

Descriptor Length: Specifies the length in bytes of the Descriptor Value.

4.2.43.2.2 Tagging Administrative Status Descriptor

The format of the Tagging Administrative Status descriptor is shown in table 124.

Table 124 – Tagging Administrative Status Descriptor

Bits / Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	Descriptor Control = 01h	Descriptor Type = 01h	Descriptor Length = 0004h	
1	Administrative Tagging Mode			

The defined Administrative Tagging Modes are shown in table 125.

Table 125 – Administrative Tagging Modes

Value	Notation	Description
0000 0001h	OFF	The FC_Port shall not perform VFT Tagging
0000 0002h	ON	The FC_Port may perform VFT Tagging if the peer does not prohibit it
0000 0003h	AUTO	The FC_Port may perform VFT Tagging if the peer request it

In absence of any explicit configuration, the default Administrative Tagging Mode of a VF capable N_Port or F_Port should be AUTO.

Table 126 shows how VFT tagging is negotiated between peer FC_Ports.

Table 126 – Tagging Mode Negotiation

		Peer Tagging Mode		
		OFF	**ON**	**AUTO**
Local Tagging Mode	**OFF**	Non Tagging	Non Tagging	Non Tagging
	ON	Non Tagging	Tagging	Tagging
	AUTO	Non Tagging	Tagging	Non Tagging

4.2.43.2.3 Port VF_ID Descriptor

The format of the Port VF_ID descriptor is shown in table 127.

Table 127 – Port VF_ID Descriptor

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	Descriptor Control = 01h	Descriptor Type = 02h	Descriptor Length = 0004h	
1	Port Flags		Port VF_ID	

Port Flags: Reserved. Shall be set to zero.

Port VF_ID: The 12 least significant bit of this field shall be set to the Port VF_ID. The four most significant bit shall be set to zero. In absence of any explicit configuration, the value 001h should be used as Port VF_ID.

4.2.43.2.4 Locally-Enabled VF_ID List Descriptor

The format of the Locally-Enabled VF_ID List descriptor is shown in table 128.

Table 128 – Locally-Enabled VF_ID List Descriptor

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	Descriptor Control = 01h	Descriptor Type = 03h	Descriptor Length = 0200h	
1	VF_ID Bitmap			
..				
128				

VF_ID Bitmap: Each Virtual Fabric is identified by a bit in the VF_ID Bitmap. The high-order bit represents VF_ID zero, each successive bit represents the successive VF_ID, and the low-order bit represents VF_ID 4095. Virtual Fabric K is allowed on the Interconnect_Port if the Kth bit of the VF_ID Bitmap is set to one; is disallowed if the Kth bit of the VF_ID Bitmap is set to zero. The bit representing the Control VF_ID (see FC-FS-2) shall be set to zero.

The list of Virtual Fabrics operational over a link is computed by performing a bit-wise 'AND' between the received VF_ID Bitmap and the locally configured VF_ID Bitmap.

4.2.43.2.5 Vendor Specific Descriptor

The format of the Vendor Specific descriptor is shown in table 129.

Table 129 – Vendor Specific Descriptor

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	Descriptor Control	Descriptor Type	Descriptor Length	
1	T10 Vendor ID			
2				
3	Vendor Specific			
..				
N				

T10 Vendor ID: Shall be set to the Vendor's T10 Vendor ID.

4.2.43.3 EVFP_COMMIT Message Payload

Both EVFP_COMMIT Request and EVFP_COMMIT Accept have no Message Payload.

4.2.44 Link Keep Alive (LKA)

4.2.44.1 Overview

The LKA ELS is used for traffic generation. It provides a means to generate traffic in order to confirm that the link is still intact and/or to ensure the link is not terminated due to lack of traffic. The LKA ELS was specifically designed to keep Fibre Channel backbone links alive (e.g., some TCP implementations will disconnect connections that are not used for some time period).

The LKA ELS is sent by a VE_Port or B_Access portal to a remote peer in order to determine the health of a link between them, or simply to generate traffic to keep a link from being terminated. Should a link be comprised of more than one physical or virtual connection, the LKA may be transmitted on each of the connections. If a connection is configured to handle only specific class(es) of traffic, the LKA shall be sent on a class of service the connection is configured for.

The LKA ELS request Sequence shall consist of a single frame requesting the recipient to reply using the ACC reply Sequence consisting of a single frame. The LKA ELS request frame shall indicate End_Sequence and Sequence Initiative transfer as well as other appropriate F_CTL bits as defined in FC-FS-2. The LKA ELS shall be transmitted as a single frame Sequence and the ACC reply Sequence is also a single frame Sequence. The LKA ELS shall be transmitted as an Exchange that is separate from any other Exchange. The LKA ELS is applicable to Class F, 2, 3, and 4.

The LKA ELS may be sent at any time. The LKA ELS should be sent at least every K_A_TOV if no traffic has been sent and/or received on the connection. The default value for K_A_TOV shall be 1/2 E_D_TOV.

If an accept is not received within E_D_TOV, a new LKA ELS may be transmitted in a new Exchange. The Exchange used for the previous LKA request shall be aborted.

Upon discovering an error (e.g. due to service reject or failure to receive a timely accept in response to one or more LKA ELS requests), the initiator shall initiate appropriate exception handling. The definition of appropriate exception handling is topology-specific.

4.2.44.2 Protocol

a) Link Keep Alive request Sequence; and

b) LS_ACC or LS_RJT reply Sequence.

4.2.44.3 Request Sequence

Addressing: The S_ID field shall be set to FFFFFDh, indicating the Fabric Controller of the VE_Port or B_Access portal originating the request. The D_ID field shall be set to FFFFFDh, indicating the Fabric Controller of the remote peer.

Payload: The format of the LKA Request Payload is shown in table 130.

Table 130 – LKA Payload

Bits / Word	31... 24	23... 16	15... 08	07... 00
0	80h	00h	00h	00h

4.2.44.4 Reply Sequence

LS_RJT: LS_RJT signifies rejection of the LKA request.

LS_ACC: LS_ACC signifies that the connection is intact. The format of the LS_ACC payload is found in table 131.

Table 131 – LKA LS_ACC Payload

Bits Word	31... 24	23... 16	15... 08	07... 00
0	02h	00h	00h	00h

4.2.45 Define FFI Domain Topology Map (FFI_DTM)

4.2.45.1 Description

The FFI_DTM ELS Request shall transfer a complete initial or replacement Domain Topology Map to the Domain Controller of the AE Principal Switch.

Support for this ELS is mandatory for a Domain Controller in an AE Principal Switch or an AE Switch that is capable of becoming an AE Principal Switch. Support for this ELS is optional for Nx_Ports.

If the Domain Controller that receives an FFI_DTM request currently is not the AE Principal Switch, it shall respond with an LS_RJT reply with a reason code of “Logical error”.

If a destination receives this ELS Request that is not a Domain Controller, it shall respond with an LS_RJT reply with a reason code of “Command not supported”.

4.2.45.2 Protocol

a) FFI Domain Topology Map Request Sequence; and

b) LS_ACC or LS_RJT Reply Sequence.

4.2.45.3 Request Sequence

Addressing: The S_ID designates the Nx_Port sending an FFI Domain Topology Map. The D_ID designates the Domain Controller of the AE Principal Switch in the form FFFCxxh, where xx is the one-byte value assigned to the Domain Controller.

Payload: The format of the FFI_DTM Request Payload is shown in table 132.

Table 132 – FFI_DTM Payload

Bits / Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	FFI_DTM (A0h)	00h	Payload Length	
1	FFI Incarnation Number			
2	Reserved	Reserved	Number of FFI Link State Records	
3	FFI Link State Records			
..				
n				

Payload Length: This field is the length in bytes of the entire Payload, inclusive of the length of word 0. This value shall be a multiple of 4. The minimum value of this field is 28. The maximum value of this field is 65532.

FFI Incarnation Number: This field contains the new incarnation of the FFI Domain Topology Map.

Number of FFI Link State Records: This field shall specify the number of FFI Link State Records that follow this field. The minimum value is 2.

FFI Link State Records: This field contains all of the individual FFI Link State Records that describe the Domain Topology Map of the Avionics Fabric. The format of the FFI Link State Record is described in FC-SW-4.

4.2.45.4 Reply Sequence

LS_RJT: LS_RJT signifies the rejection of the FFI_DTM Request. The LS_RJT reply contains an appropriate reject reason code.

LS_ACC: LS_ACC signifies acceptance of the FFI_DTM Request. The format of the LS_ACC Payload for FFI_DTM is shown in table 133.

Table 133 – FFI_DTM LS_ACC Payload

Bits / Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	02h	00h	00h	00h

4.2.46 Request FFI Domain Topology Map (FFI_RTM)

4.2.46.1 Description

The FFI_RTM ELS Request shall request the Domain Controller of the AE Principal Switch to return the current Domain Topology Map in the LS_ACC Reply Sequence.

Support for this ELS is mandatory for a Domain Controller in an AE Principal Switch or an AE Switch that is capable of becoming an AE Principal Switch. Support for this ELS is optional for Nx_Ports.

If the Domain Controller that receives an FFI_RTM request currently is not the AE Principal Switch, it shall respond with an LS_RJT reply with a reason code of “Logical error”.

If a destination receives this ELS Request that is not a Domain Controller, it shall respond with an LS_RJT reply with a reason code of “Command not supported.”

4.2.46.2 Protocol

a) FFI Domain Topology Map Request Sequence; and

b) LS_ACC or LS_RJT Reply Sequence.

4.2.46.3 Request Sequence

Addressing: The S_ID field value identifies the Nx_Port requesting the Domain Topology Map from the Domain Controller of the AE Principal Switch. The D_ID designates the Domain Controller of the AE Principal Switch in the form FFFCxxh, where xx is the one-byte value assigned to the Domain Controller.

Payload: The format of the FFI_RTM Request Payload is shown in table 134.

Table 134 – FFI_RTM Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	FFI_RTM (A1h)	00h	00h	00h

4.2.46.4 Reply Sequence

LS_RJT: LS_RJT signifies the rejection of the FFI_RTM Request. The LS_RJT reply contains an appropriate reject reason code.

LS_ACC: LS_ACC signifies acceptance of a valid FFI_RTM Request and that the AE Principal Switch has transmitted the requested data. The format of the LS_ACC Payload for FFI_RTM is shown in table 135.

Table 135 – FFI_RTM LS_ACC Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	02h	00h	Payload Length	
1	FFI Incarnation Number			
2	Reserved	Reserved	Number of FFI Link State Records	
3	FFI Link State Records			
..				
n				

Payload Length: This field is the length in bytes of the entire Payload, inclusive of the length of word 0. This value shall be a multiple of 4 bytes. The minimum value of this field is 28 bytes. The maximum value of this field is 65532.

FFI Incarnation Number: This field contains the current incarnation of the FFI Domain Topology Map.

Number of FFI Link State Records: This field shall specify the number of FFI Link State Records that follow this field. The minimum value is 2.

FFI Link State Records: This field contains all of the individual FFI Link State Records that describe the Domain Topology Map of the Avionics Fabric. The format of the FFI Link State Record is described in FC-SW-4.

4.2.47 FFI AE Principal Switch Selector (FFI_PSS)

4.2.47.1 Description

The FFI_PSS ELS Request shall be sent to an AE Switch that is not currently the AE Principal Switch in order to command the recipient to become the AE Principal Switch.

Support for this ELS is mandatory for a Domain Controller in an AE Principal Switch or an AE Switch that is capable of becoming an AE Principal Switch. Support for this ELS is optional for Nx_Ports.

If the Domain Controller that receives an FFI_PSS request currently is the AE Principal Switch or is not capable of becoming the AE Principal Switch, it shall respond with an LS_RJT reply with a reason code of “Logical error”.

If a destination receives this ELS Request that is not a Domain Controller, it shall respond with an LS_RJT reply with a reason code of “Command not supported”.

4.2.47.2 Protocol

a) FFI AE Principal Switch Selector Request Sequence; and

b) LS_ACC or LS_RJT Reply Sequence.

4.2.47.3 Request Sequence

Addressing: The S_ID field value identifies the Nx_Port requesting a change of AE Principal Switch. The D_ID field value identifies the Domain Controller of an AE Switch that is not the current AE Principal Switch but is capable of becoming an AE Principal Switch in the form FFFCxxh, where xx is the one-byte value assigned to the Domain Controller.

Payload: The format of the FFI_PSS Request Payload is shown in table 136.

Table 136 – FFI_PSS Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	FFI_PSS (A2h)	00h	00h	00h

4.2.47.4 Reply Sequence

LS_RJT: LS_RJT signifies the rejection of the FFI_PSS Request. The LS_RJT reply contains an appropriate reject reason code.